Organic molecules and functional groups

1. recall that there is a family of hydrocarbons called alkanes;
2. recall the names and molecular formulae of the alkanes methane, ethane, propane and butane;
3. translate between molecular, structural and ball-and-stick representations of simple organic molecules;
4. recall that alkanes burn in plenty of air to give carbon dioxide and water;
5. understand that alkanes are unreactive towards aqueous reagents because C—C and C—H bonds are unreactive.
6. **represent chemical reactions by balanced equations, including state symbols.**

Alcohols

7. recall the names and molecular formulae of methanol and ethanol;
8. recall two uses of methanol and two of ethanol;
9. recognise the formulae of alcohols;
10. understand that the characteristic properties of alcohols are due to the presence of the –OH functional group;
11. recall how ethanol compares in its physical properties with water and with alkanes;
12. understand that alcohols burn in air because of the presence of a hydrocarbon chain;
13. **recall the reaction of alcohols with sodium and how this compares with the reactions of water and alkanes with this metal.**

Carboxylic acids

14. understand that the characteristic properties of carboxylic acids are due to the presence of the –COOH functional group;
15. recall the names and formulae of methanoic and ethanoic acids;
16. recognise the formulae of carboxylic acids;
17. recall that many carboxylic acids have unpleasant smells and tastes and are responsible for the smell of sweaty socks and the taste of rancid butter;
18. understand that carboxylic acids show the characteristic reactions of acids with metals, alkalis and carbonates;
19. recall that vinegar is a dilute solution of ethanoic acid.

Esters

20. understand that carboxylic acids react with alcohols, in the presence of a strong acid catalyst, to produce esters;
21. write a word equation for the formation of an ester;
22. recall that esters have distinctive smells;
23. recall that esters are responsible for the smells and flavours of fruits;
24. recall the use of esters in products such as food, perfumes, solvents and plasticisers;
25. **understand the procedure for making an ester (such as ethyl ethanoate) from a carboxylic acid and an alcohol;**
26. **understand techniques used to make a liquid ester including heating under reflux, distillation, purification by treatment with reagents in a tap funnel and drying;**
27. understand that fats are esters of glycerol and fatty acids;
28. recall that living organisms make fats and oils as an energy store;
29. recall that animal fats are mostly saturated molecules and that vegetable oils are mostly unsaturated molecules;
30. understand that in saturated compounds all the C—C bonds are single but that in unsaturated compounds there are C=C double bonds.